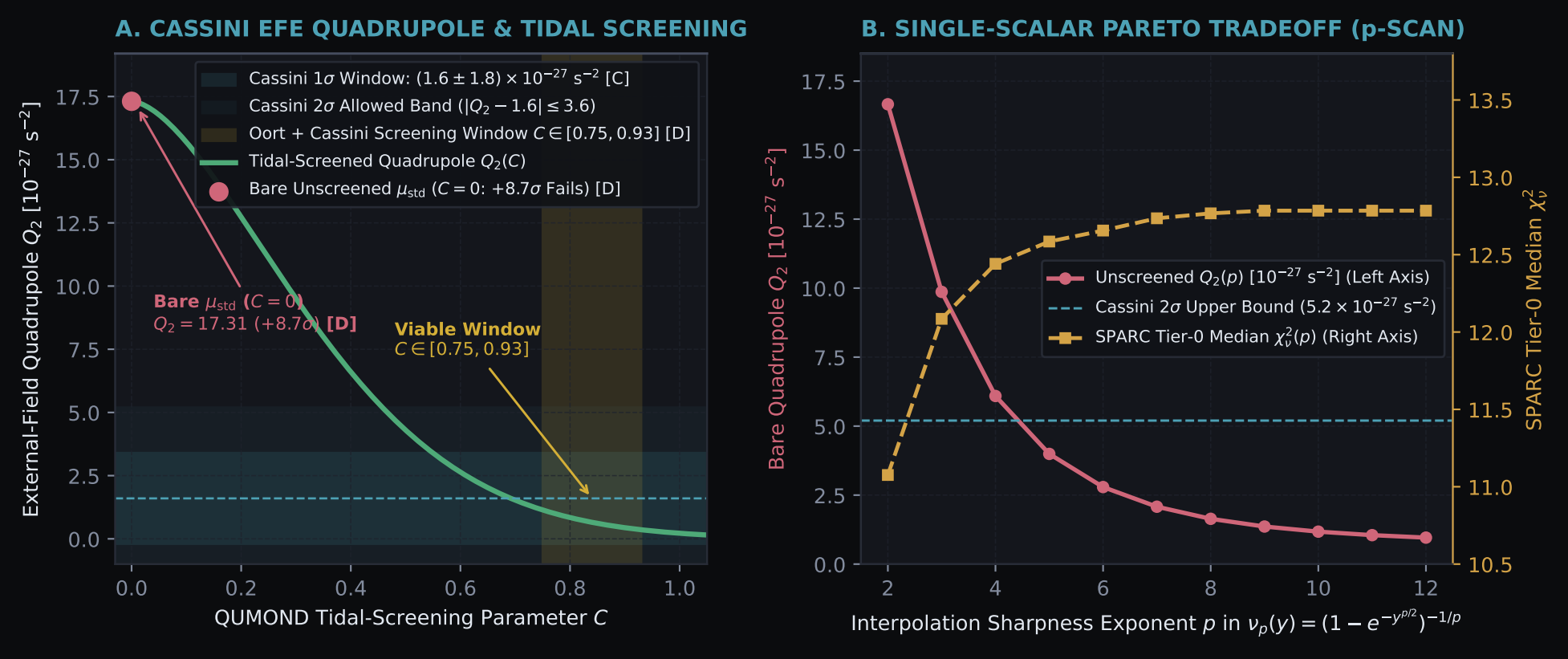
A. CASSINI EFE QUADRUPOLE & TIDAL SCREENING
Cassini 1σ Window: (1.6 ± 1.8) × 10⁻²⁷ s⁻² [C]
Cassini 2σ Allowed Band (|Q₂ − 1.6| ≤ 3.6)
Oort + Cassini Screening Window C ∈ [0.75, 0.93] [D]
Tidal-Screened Quadrupole Q₂(C)
Bare Unscreened μ_std (C = 0: +8.7σ Fails) [D]
Bare μ_std (C = 0)
Q₂ = 17.31 (+8.7σ) [D]
Viable Window
C ∈ [0.75, 0.93]
External-Field Quadrupole Q₂ [10⁻²⁷ s⁻²]
QUMOND Tidal-Screening Parameter C
B. SINGLE-SCALAR PARETO TRADEOFF (p-SCAN)
Unscreened Q₂(p) [10⁻²⁷ s⁻²] (Left Axis)
Cassini 2σ Upper Bound (5.2 × 10⁻²⁷ s⁻²)
SPARC Tier-0 Median χ²_ν(p) (Right Axis)
Bare Quadrupole Q₂ [10⁻²⁷ s⁻²]
SPARC Tier-0 Median χ²_ν
Interpolation Sharpness Exponent p in ν_p(y) = (1 − e^{−y^{p/2}})^{−1/p}